files are overwritten, which makes them unrecoverable. Also, you cannot say for sure that all disks and file systems are supported.

Backing up is a process that you cannot think of when your data is lost. You should take backups of your files/folders when you make important changes. Many backup utilities can automate this process, making things easier. Copies of your data (or of only the changes) can be stored in a remote location, or just in another local disk, making your data 'immortal'. Restoring from backups is a simple process.

Why a simple copy-paste isn't enough

Does the term 'backup' sound challenging to you? For many, it seems to be a process that can only be done by experts with the help of sophisticated tools. But let me correct that—just keeping a copy of your files in a CD or some other storage media is a backup. If you are an experienced user, you might think that I am silly to mention this as an important point. But believe me, there are many to whom data backup is still an alien concept.

However, a simple copy-paste isn't enough, especially when you are doing a long-term project. Imagine that you are shooting a movie. Every day you capture some video clips and add them to your library (say, a folder titled 'CLIPS'). What should be your backup strategy here? You have two choices, if you don't want to think about the backup software at this stage. The first one is to take the copy of the folder 'CLIPS' each time you make a change, and rename it with the current date (e.g., CLIPS_28_07_2015). The second one is to take copies of the changes only.

In the first case, you are wasting a lot of space, while in the second, it is very difficult to keep track of each change. Either way, you have to do everything manually, wasting a lot of time and making way for confusion. Therefore, automated backup utilities are necessary.

Online or offline backups?

We have just seen the importance of automated backup utilities, and we are going to check two of them. But we should discuss another important thing before that—whether automatic or manual backups are best for you. Also, should you go for CDs, DVDs, hard disks or online storage? And further, we need to think about how to preserve these backups.

The answers are based on the longevity and reliability of these media. No medium has an infinite lifespan. Their actual lifespan may be topics of considerable debate and depend on a lot of parameters. But studies show that you cannot rely upon CDs and DVDs for periods of over one or two decades while hard disks have an amazingly short life span of just four-five years. It will be interesting to know more about M-Disc, which is supposed to have a life span of 1000 years. But it doesn't seem to be feasible right now. And I don't think tapes will be of any interest to a desktop user.

This means that if you rely on a local storage device, you have to constantly monitor and duplicate it. There is also another reason to do this—present data storage standards will become obsolete as technology advances.

This is where the importance of cloud storage comes in. Rely upon some online storage provider or buy some space from a hosting company so that they will take care of your backups. This means that you take the backup of your local files and upload it to a server, and your service provider will do everything to keep it forever. With remote storage, you may need encryption to keep your data safe. Thanks to the tools we are going to study, all of this is now automated.

Déjà Dup, the beginner's tool

Déjà Dup isn't a tool for novices because of how simple it is; rather, it is a highly automated program that is really user-friendly, which makes it suitable for a beginner. In fact, it is a clever graphical wrapper around Duplicity, which is a sophisticated command line utility. We'll be having a look at Duplicity later.

Since Déjà Dup is a front-end for Duplicity, it also follows the 'chain method', whereby a full backup is only taken the first time. The backups that follow note the changes only, which is a tricky method used to save space. It is recommended to take full backups periodically so that you don't have to be afraid of any bugs.

Déjà Dup is shipped with Ubuntu while the basic editions of other distros may not have it. You can install it using your default package manager. In Debian-like systems, the following command will do (open the terminal, type it, and press *Enter*):

Figure 1: Déjà Dup in Debian 8

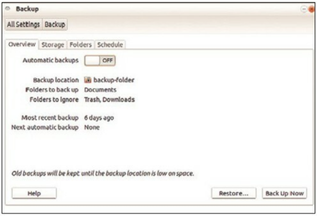

Figure 2: Déjà Dup in Ubuntu 12.04